INCH-POUND
MIL-DTL-24643/28F
1 October 2009
SUPERSEDING
MIL-DTL-24643/28E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS1S50MU (INCLUDING VARIATION LS1S50MUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS1S50MU and LS1S50MUS (double overall shield).

Construction, Non-Watertight

First	-	Copper conductor, tin-coated, ASTM B286, Size 22 – 7strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, white. Nominal wall thickness of 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth	-	Braided shield of 36 AWG, tin-coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by Method 2 on the inner tape.
Sixth	-	The required number of singles (see table I or II) cabled with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with no. 1. in the center outward. Fillers may be used as necessary to make a firm, well-rounded assembly.
Seventh	-	An optional binder.

Construction, Variant for Type LS1S50MU

Eighth	-	Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

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Construction, Variant for Type LS1S50MUS

Eighth	-	Braided shield of tin-coated copper with a minimum coverage of 85 percent.
Ninth	-	Spirally wrapped 0.002-inch thick polyethylene (polyester film) separator tape with 25 percent minimum lap.
Tenth	-	Braided shield of tin-coated copper with a minimum coverage of 85 percent.
Eleventh	-	Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

TABLE I. Details (unshielded).

Military part no. M24643/28	Type and size	Number of conductors	Diameter over core nom. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UN	LS1S50MU-16	16	0.655	0.760	0.825
-02UN	LS1S50MU-20	20	0.760	0.835	0.905
-03UN	LS1S50MU-40	40	1.015	1.095	1.185
-04UN	LS1S50MU-70	70	1.380	1.465	1.555

TABLE II. Details (shielded).

Military part no. M24643/28	Type and size	Number of conductors	Diameter over core (inch) (nom.)	Overall diameter (inch)	
				min.	max.
-01UD	LS1S50MUS-16	16	0.655	0.820	0.885
-02UD	LS1S50MUS-20	20	0.760	0.895	0.965
-03UD	LS1S50MUS-40	40	1.015	1.155	1.245
-04UD	LS1S50MUS-70	70	1.380	1.525	1.615

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	17.54
Voltage withstand (volts, root mean square, min.)	
Conductor to shield	5000
Component shield to component shield	500
Overall shield to overall shield (Type LS1S50MUS only)	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to shield	2500
Component shield to component shield	100

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Overall shield to overall shield (Type LS1S50MUS only)	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws (Type LS1S50MUS only)	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance, at 1 megahertz (MHz) (ohms)	50±2
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Attenuation, at 400 MHz (decibel (db)/100 ft max.)	18
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield (conformance to material, construction and coverage)	No failure
Surface transfer impedance (Type LS1S50MUS only)	

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Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS1S50MU and LS1S50MUS - all sizes	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-032)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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